

## Paper 12 §3 — The NP-Pump Nonlinearity: Dynamics on the Paper 9 Schur Circle (v3)

**Draft version:** v3 (2026-05-06, ~03:45 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** Paper 12 §3 v1 + Model Council rigor pass (Opus 4.7) + boundary-KKT interior-sup audit (paper12\_q5\_boundary\_kkt\_interior\_sup.py, 2026-05-06 01:45 PDT). **Revision summary (vs v2):** Phase-2 polish based on Opus's v2 residuals (RES-3 Filippov  $\lambda$  existence condition explicit; Remark 3.4.3 sharp-Heaviside apologia trimmed). Audit-runtime details consolidated with verification-files block.

**Revision summary (vs v1, retained from v2):** All 11 v1 issues from Opus 4.7 review applied. Three load-bearing changes (R1 Theorem 3.6.1 sign+framing fix forced by 24/28 boundary-KKT seeds where interior-sup < boundary; R2 Theorem 3.4.1 restated with Sard-style transversality and renamed Conditional Regularity; S3-7 escape-into-interior framing replaced by escape-into-domain-interior qualified by audited gain measurements). Plus 8 surgical fixes.

**Pillar-delivery signpost (per GPT-5.5 council finding):** §3 delivers Pillar 1 (Paper 9 dyadic extension with NP-pump) and the mathematical core of Pillar 3 (audit-grounded substrate, via the event-indexed  $F_{21}$ -module  $S_k$  in §3.7). Pillars 2 (dual-substrate residue) and 4 (Fifth Paradigm legitimization) are not delivered by §3 in isolation; Pillar 2 routes through §4 (commensurability hierarchy) and Pillar 4 routes through §1 / §5.

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### 3. The NP-Pump Nonlinearity

#### 3.1 Setup and inheritance from Paper 9

Throughout §3,  $V := V^{14}$  denotes the  $G_2$  adjoint representation and  $W := V \oplus V$  the joint-state space of the dyad, with the diagonal  $G_2$ -action  $g \cdot (x, y) = (g \cdot x, g \cdot y)$ . Paper 9 v1.3.3 [] establishes three results we take as given:

- **(P9.I3.6.1) Schur uniqueness of the isometric coupling.** The equivariant isometry group of  $W$  under the diagonal  $G_2$ -action is  $SO(2)$ , generated by

$$\Psi_\theta = I_V \otimes R_\theta = \begin{pmatrix} \cos \theta I_V & -\sin \theta I_V \\ \sin \theta I_V & \cos \theta I_V \end{pmatrix},$$

with infinitesimal generator

$$J = I_V \otimes \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -I_V \\ I_V & 0 \end{pmatrix}.$$

- **(P9.T9.4) Existence and real-analyticity of the joint fixed point.** For  $\theta \in \Omega_\theta := \{\theta \in (0, \pi/2) : \det M(\theta) \neq 0\}$ , where

$$M(\theta) = \begin{pmatrix} I - r \cos \theta R_A & r \sin \theta R_A \\ -r \sin \theta R_B & I - r \cos \theta R_B \end{pmatrix},$$

the joint fixed point

$$\hat{z}(\theta) := M(\theta)^{-1}b, b := (b_A; b_B),$$

exists, is unique, and is real-analytic on  $\Omega_\theta$ .

- **(P9.P9.5) Conditional gain in the min-per-observer functional.** At bias-alignment  $\varphi=0$  and  $r_A=r_B=r \in (0, 6/7)$ , the functional  $c(\theta) := C_{\min}(\hat{z}(\theta))$  where  $C_{\min} = \min(C_A, C_B)$  admits a non-empty improvement region over the decoupled value  $c_{dec} := c(0)$  for Haar-positive-measure  $(R_A, R_B)$ , with non-universal optimum  $\theta^*$  (Conjecture 9.5', 50-seed Monte Carlo verification in Appendix V.3.i).

Paper 9 further proves (Note 9.6) that in the *linear* regime the joint rate is locked at  $\max(r_A, r_B)$  for all  $\theta$  (Theorem 9.2, Corollary 9.3, Schur's lemma). Rate improvement therefore requires departing from the linear regime. The present section constructs that departure in a way that (i) preserves the Schur-circle geometry of (P9.L3.6.1) exactly, (ii) produces piecewise-real-analytic trajectories on parameter sets of full Lebesgue measure, and (iii) respects the closed-interval domain  $\theta \in [0, \pi/2]$  of Paper 9.

### 3.2 The canonical tear direction

**Notational convention.** We work branch-wise throughout. Let

$$c_A(\theta) := C_A(\hat{z}(\theta)), c_B(\theta) := C_B(\hat{z}(\theta)),$$

and  $c(\theta) := \min(c_A(\theta), c_B(\theta))$ . The active branch  $\sigma(\theta) \in \{A, B\}$  is  $A$  when  $c_A < c_B$ ,  $B$  when  $c_B < c_A$ , and undefined ("tie") on  $\Sigma_{\min} := \{\theta : c_A(\theta) = c_B(\theta)\}$ . The sign-degeneracy locus per branch is

$$Z_\sigma := \{\theta \in \Omega_\theta : c_\sigma'(\theta) = 0\}, Z := Z_A \cup Z_B.$$

On  $\Sigma_{\min}$ ,  $c$  is continuous but its classical derivative is not single-valued; we use the Clarke generalized gradient on this locus.

**Definition 3.2.1 (Schur-derived tear direction).** On  $Z^c := \Omega_\theta \setminus Z$  where the active-branch derivative is single-signed and nonzero, the *canonical interior tear direction* is

$$\xi_\star^{int}(\theta) := \text{sgn}(c_{\sigma(\theta)}'(\theta)) \cdot \partial_\theta,$$

with push-forward to the joint-fixed-point manifold

$$\Xi_\star^{int}(\theta) := \text{sgn}(c_{\sigma(\theta)}'(\theta)) \cdot \hat{z}'(\theta), \hat{z}'(\theta) = -M(\theta)^{-1}M'(\theta)\hat{z}(\theta).$$

On boundary points  $\theta \in [0, \pi/2]$ , the tear direction is projected into the inward-pointing tangent cone  $T_{[0, \pi/2]}(\theta)$ :

$$\xi_\star(\theta) := \Pi_{T_{[0, \pi/2]}(\theta)}(\xi_\star^{int}(\theta)).$$

On  $Z \cup \Sigma_{\min}$ ,  $\xi_\star$  is left set-valued, with canonical element determined (if present) by the first non-zero odd derivative of  $c_\sigma$  at  $\theta$  on  $Z$ , or by the Clarke generalized gradient on  $\Sigma_{\min}$ .

**Proposition 3.2.2 (Line uniqueness from Schur).** *Any  $G_2$ -equivariant isometric infinitesimal direction along the Paper 9 Schur fiber lies in the one-dimensional space  $R\partial_\theta$ , with generator  $J$  acting on  $W$  via  $d\Psi_\theta/d\theta = J\Psi_\theta$ .*

*Proof.* The equivariant isometry group of  $W$  is  $SO(2)$  by Paper 9 Lemma 3.6.1; its Lie algebra is the one-dimensional  $RJ$ .  $\square$

**Axiom 3.2.3 (Coherence-ascent design principle).** *Define the  $\xi_\bullet$ -sign on  $Z^c$  by  $\text{sgn}(c'_\sigma(\theta))$  in the interior, and project into  $T_{[0,\pi/2]}(\theta)$  at boundary points.*

The line-uniqueness of Prop 3.2.2 is a consequence of  $G_2$ -equivariance + isometry + Paper 9 orientation; the sign is *not*. Both  $+\partial_\theta$  and  $-\partial_\theta$  are admissible directions under equivariance and isometry alone. The choice of  $\text{sgn}(c'_\sigma)$  is an additional design axiom — the *coherence-ascent principle* — that selects the direction in which the augmented dynamics climbs  $c_\sigma$  rather than descending it. We state this as an explicit modeling axiom (3.2.3) rather than a derived consequence.

**Remark 3.2.4 (Boundary projection vs interior sign).** At boundary-left KKT points, where  $c'_\bullet(0) \leq 0$  (Definition 3.5.1 below), the interior-sign rule of 3.2.3 gives  $\text{sgn}(c'_\bullet(0)) \in \{-1, 0\}$ , so the un-projected tear  $\xi_\bullet^{int}$  would push to  $\theta < 0$  outside the admissible domain  $[0, \pi/2]$ . The tangent-cone projection  $\Pi_{T_{[0,\pi/2]}(0)} = \cdot$  truncates this to  $\xi_\bullet(0) = 0$  — i.e., *no escape from the boundary along the linear  $\xi_\bullet$  direction*. This is the load-bearing correction relative to the v1 draft, which mistakenly admitted unprojected sign-driven escape at boundary-KKT points; the implications for Theorem 3.6.1 are made precise in §3.6 below.

**Remark 3.2.5 (Nonlinear inherits linear geometry).** Definition 3.2.1 asserts that the nonlinear dynamics of Paper 12 lives on *the same* Schur circle as Paper 9's linear coupling  $\Psi_\theta$ . The NP-pump does not generate a new geometric object; it generates a new dynamics on the existing one-parameter family  $\{\Psi_\theta : \theta \in [0, \pi/2]\}$ . The only Schur-free parameter in the nonlinear theory is the jump length  $\rho_i > 0$  at each firing event, determined by the NP amplitude rather than  $G_2$ -representation theory.

### 3.3 The augmented flow and event stratification

**Definition 3.3.1 (NP amplitude and paradox mass).** For each branch  $\sigma \in \{A, B\}$  let

$$p_\sigma(\theta) := \dot{c}_\sigma \partial_\theta F_\sigma(\theta) \dot{c}_\sigma^2, F_\sigma(\theta) := -c_\sigma(\theta).$$

We identify  $p_\sigma$  with the squared *paradox mass* of the Paper 12 thesis memo: a non-negative, branch-wise real-analytic quantity that accumulates wherever the coherence functional changes rapidly. The squared-norm form is chosen (over the raw norm) to preserve analyticity through zeros of  $F'_\sigma$ . The *NP amplitude* is

$$a_\sigma(\theta) := \alpha \cdot p_\sigma(\theta), \alpha > 0.$$

The *coherence gap* is  $g_\sigma(\theta) := C_{crit} - c_\sigma(\theta)$  with  $C_{crit} \in (0, 1)$ . The firing predicate is

$$NP_{\sigma}(\theta) := 1 \{g_{\sigma}(\theta) \geq 0, a_{\sigma}(\theta) \geq NP_{crit}\},$$

with  $NP_{crit} > 0$  a fixed amplitude threshold.

**Definition 3.3.2 (Augmented flow).** On the active branch of  $[0, \pi/2] \setminus \Sigma_{tot}$  (with  $\Sigma_{tot}$  as in Definition 3.3.3), the augmented flow is

$$\dot{\theta} = \Pi_{T_{[0, \pi/2]}(\theta)} \left[ -\eta F'_{\sigma}(\theta) + \kappa NP_{\sigma}(\theta) \xi_{\sigma}(\theta) \right], \eta, \kappa > 0,$$

where  $\Pi_{T_{[0, \pi/2]}(\theta)}$  is projection onto the closed- interval tangent cone (Definition 3.5.2). At switching events on  $\Sigma_{min}$ , the flow is treated as a Filippov differential inclusion with the equivalent-control protocol of (3.4.A) below; at crossings of  $\Sigma_{\Theta}$ , as a Carathéodory solution; at NP firings on  $G_{NP}$ , as a hybrid jump (Definition 3.3.4).

**Definition 3.3.3 (Total event stratification).** The nonsmooth set for the augmented flow is the union

$$\Sigma_{tot} = \Sigma_{min} \cup \Sigma_{\Theta} \cup Z \cup G_{NP} \cup \partial[0, \pi/2],$$

where  $\Sigma_{min}, \Sigma_{\Theta}, Z$  are as defined above,  $\partial[0, \pi/2] = \{0, \pi/2\}$  is the closed-interval boundary, and  $G_{NP} := \bigcup_{\sigma} (G_{\sigma}^{amp} \cup G_{\sigma}^{coh})$  is the *firing manifold*, split into amplitude- and coherence-triggered entry submanifolds (Definition 3.3.5).

**Definition 3.3.4 (Hybrid jump at firing events).** At a firing event  $\theta^- \in \Omega_{\theta} \setminus (Z \cup \partial[0, \pi/2])$  with  $(\theta^-, \hat{z}(\theta^-)) \in G_{NP}$  on branch  $\sigma$ , the jump map is

$$\theta^{+i} = \theta^- + \rho_{\sigma}(\theta^-) \text{sgn}(c_{\sigma}'(\theta^-)), i$$

with  $\rho_{\sigma}: G_{NP} \rightarrow \mathbb{R}_{>0}$  a real-analytic jump-length function depending on the NP-amplitude surplus  $a_{\sigma}(\theta^-) - NP_{crit}$ . To prevent Zeno accumulation, we require the Zeno-avoidance condition  $J_{\sigma}(D_{\sigma}) \subset \Gamma \setminus D_{NP}$ , i.e., the jump map lands outside the firing region  $D_{NP} = \bigcup_{\sigma} \{(\theta, \hat{z}(\theta)) : g_{\sigma} \geq 0, a_{\sigma} \geq NP_{crit}, c_{\sigma}' \neq 0\}$ ; if not, we add a refractory dwell  $\tau_{ref} > 0$  between consecutive firings.

**Definition 3.3.5 (Amplitude vs coherence entry manifolds).** The firing manifold  $G_{NP}$  decomposes into two entry submanifolds of  $D_{NP}$  along the unperturbed flow:

- *Amplitude-triggered:*  $G_{\sigma}^{amp} := \{g_{\sigma} > 0, a_{\sigma} = NP_{crit}, c_{\sigma}' \neq 0, L_f a_{\sigma} > 0\}$  — the NP amplitude crosses the critical value while the coherence is already below  $C_{crit}$ .
- *Coherence-triggered:*  $G_{\sigma}^{coh} := \{g_{\sigma} = 0, a_{\sigma} \geq NP_{crit}, c_{\sigma}' \neq 0, L_f g_{\sigma} > 0\}$  — the coherence gap  $g_{\sigma}$  crosses zero *increasing* (so coherence drops below  $C_{crit}$ ) while the NP amplitude is already supercritical.

Here  $L_f$  denotes the Lie derivative along the unperturbed gradient flow  $f(\theta) = -\eta F'_{\sigma}(\theta)$ . The transversality conditions  $L_f a_{\sigma} > 0$  and  $L_f g_{\sigma} > 0$  ensure the system genuinely enters the firing

region rather than grazing it; tangential contact points belong to the exceptional grazing stratum.

### 3.4 Branchwise real-analyticity and the Conditional Regularity Theorem

The Filippov sliding-mode protocol on  $\Sigma_{min}$  admits three regimes determined by the branch fields  $f_A, f_B$  and the kink-normal  $D\Delta$  where  $\Delta := c_A - c_B$ :

**(3.4.A) Sliding mode.** If both one-sided fields point inward to  $\Sigma_{min}$  (i.e.,  $D\Delta \cdot f_A$  and  $D\Delta \cdot f_B$  have opposite signs), the trajectory slides along  $\Sigma_{min}$  with the equivalent-control formula

$$\dot{\theta} = \lambda f_A + (1 - \lambda) f_B, \lambda \in [0, 1] \text{ chosen so that } D\Delta \cdot \dot{\theta} = 0.$$

Solving for  $\lambda$ :

$$\lambda = \frac{D\Delta \cdot f_B}{D\Delta \cdot (f_B - f_A)}.$$

Existence of  $\lambda \in [0, 1]$  requires that the denominator  $D\Delta \cdot (f_B - f_A)$  is nonzero (this is the genericity built into (R3)) and that the resulting  $\lambda$  lies in the unit interval, which holds iff  $D\Delta \cdot f_A$  and  $D\Delta \cdot f_B$  have opposite signs as assumed. When this  $\lambda$ -existence condition fails ( $D\Delta = 0$ , or the inward-pointing condition degenerates), the contact is tangential and the trajectory is placed in the exceptional stratum (3.4.C).

**(3.4.B) Crossing mode.** If both one-sided fields point in the *same* direction across  $\Sigma_{min}$  (i.e.,  $D\Delta \cdot f_A$  and  $D\Delta \cdot f_B$  agree in sign), the trajectory transversally crosses  $\Sigma_{min}$  with no sliding; the active branch switches across the crossing.

**(3.4.C) Tangential / corner exceptional stratum.** If  $D\Delta = 0$  or one of the inward conditions degenerates, the contact is tangential and the Filippov formalism does not apply directly. Such points are placed in the exceptional stratum and excluded from the regularity theorem below.

**Theorem 3.4.1 (Conditional Regularity).** *Suppose:*

(R1)  $\hat{z} \in C^\omega(\Omega_\theta)$  and  $C_A, C_B \in C^\omega$  on an open neighborhood of  $\hat{z}(\Omega_\theta) \subset W$ ;

(R2)  $p_\sigma = |\partial_\theta F_\sigma|^2 \in C^\omega$  on each branch and  $\rho_\sigma \in C^\omega$  on  $G_{NP}$ ;

(R3) the parameter septuple  $p := (R_A, R_B, b_A, b_B, \alpha, NP_{crit}, C_{crit})$  lies in the open set  $P^\circ \subset O(N) \times O(N) \times R^{2N} \times R_{c_0}^3$  on which all guard functions  $\Delta = c_A - c_B$ ,  $g_\sigma$ ,  $a_\sigma - NP_{crit}$  and  $q_\sigma := c_\sigma'$  are transverse to zero (i.e., zero is a regular value), and the Filippov  $D\Delta$ -genericity of (3.4.A) holds where invoked;

(R4) firing times are locally finite and Definition 3.3.4's Zeno-avoidance condition holds (possibly through the refractory dwell).

Then on  $P^\circ$ , every solution of the augmented flow (Definition 3.3.2) avoiding the exceptional strata is piecewise real-analytic in  $t$ : on each open interval between events,  $\theta(t) \in C^\omega$ ; at transverse  $\Sigma_\theta$  crossings, the solution is Carathéodory; on  $\Sigma_{min}$  in regime (3.4.A), Filippov with the

equivalent-control formula; in regime (3.4.B), classical crossing; at regular firings on  $G_{NP}(E \dot{\iota}_{0,NP})$ , the jump is hybrid per Definition 3.3.4.

Furthermore,  $P^{\dot{\iota}}$  is open and dense in  $O(N) \times O(N) \times R^{2N} \times R_{\epsilon,0}^3$ .

*Proof sketch.* The genericity claim is a Sard-type argument: each guard function is real-analytic in both  $\theta$  and  $p$ , so its critical values are nowhere-dense in the codomain by the analytic-Sard theorem; the bad parameter set (where any guard fails to be transverse to zero on the trajectory) is a countable union of codimension- $\geq 1$  algebraic strata, hence nowhere dense;  $P^{\dot{\iota}}$  is its complement, open and dense by construction. On  $P^{\dot{\iota}}$ , branch-local real-analyticity of  $\theta(t)$  between events is the analytic-ODE existence theorem applied to the analytic right-hand side on  $\Omega_\theta(\Sigma \dot{\iota}_{tot})$ ; transverse guard crossings extend the solution uniquely with the Carathéodory or hybrid-jump semantics specified. Filippov sliding (3.4.A) yields a real-analytic sliding field  $f_{sl} = \lambda f_A + (1 - \lambda) f_B$  because  $\lambda$  is determined algebraically from analytic data; crossing (3.4.B) is immediate; (3.4.C) is excluded. (R4) plus locally-finite firing times bounds the number of non-smooth events on any compact  $t$ -interval.  $\square$

**Remark 3.4.2 (Why “Conditional Regularity”).** The theorem name honors that conclusions are conditional on (R1)–(R4); we additionally assert that the parameter region  $P^{\dot{\iota}}$  is open and dense in the natural ambient space, so the conditions hold for *generic* parameter choices. Verification of (R1)–(R4) at a specific parameter point is a finite-dimensional algebraic check on the guards.

**Remark 3.4.3 (Why the Heaviside is kept sharp).** Smoothing  $\Theta$  to  $\Theta_\epsilon$  produces a classical analytic ODE with no jumps, removing the “tear” character. The sharp choice is deliberate; hybrid semantics replace a classical ODE.

### 3.5 The $\kappa=0$ recovery: projected gradient flow

Setting  $\kappa=0$  in Definition 3.3.2 removes the NP-pump term and leaves the projected gradient flow on  $[0, \pi/2)$ . We now show that this unperturbed limit recovers Paper 9’s Proposition 9.5 optima as *projected* — not classical — gradient critical points.

**Definition 3.5.1 (Stratified critical point).** A point  $\theta^{\dot{\iota}} \in [0, \pi/2)$  is a *stratified critical point* of  $-c$  if at least one of the following holds:

- (Smooth interior)  $\theta^{\dot{\iota}} \in (0, \pi/2) \setminus \Sigma \dot{\iota}_{min}$  and  $c'(\theta^{\dot{\iota}}) = 0$  (Fermat);
- (Boundary-left KKT)  $\theta^{\dot{\iota}} = 0$  and  $c'_{+\dot{\iota}(0) \leq 0 \dot{\iota}} \geq 0$ ;
- (Boundary-right KKT)  $\theta^{\dot{\iota}} = \pi/2$  and  $c'_{-\pi/2} \geq 0$ ;
- ( $\Sigma_{min}$  Clarke)  $\theta^{\dot{\iota}} \in (0, \pi/2) \cap \Sigma_{min}$  and  $0 \in \text{conv}\{c_A'(\theta^{\dot{\iota}}), c_B'(\theta^{\dot{\iota}})\}$ ;
- (Corner-Clarke, exceptional)  $\theta^{\dot{\iota}} \in \{0, \pi/2\} \cap \Sigma_{min}$ , requiring the combined boundary-Clarke condition  $T_{[0, \pi/2]}(\theta^{\dot{\iota}}) \cap \text{conv}\{c_A', c_B'\}$  contains the origin in its dual cone. We treat such corner points in the exceptional stratum.

**Definition 3.5.2 (Projected gradient flow on  $[0, \pi/2]$ ).** Let  $T_{[0, \pi/2]}(\theta)$  be the closed-interval tangent cone at  $\theta$ :  $T_{[0, \pi/2]}(0) = \dot{\iota}$ ,  $T_{[0, \pi/2]}(\pi/2) = (-\infty, 0 \dot{\iota}]$ ,  $T_{[0, \pi/2]}(\theta) = R$  for  $\theta \in (0, \pi/2)$ . The projected gradient flow on  $[0, \pi/2] \setminus \Sigma_{\dot{\iota}_{\min}}$  is

$$\dot{\theta} = \Pi_{T_{[0, \pi/2]}(\theta)}(\eta c'(\theta)).$$

On  $\Sigma_{\min}$ ,  $c'$  is replaced by the Clarke generalized gradient  $\partial_C(-c)(\theta) = \text{conv}\{-c_A'(\theta), -c_B'(\theta)\}$  and the flow is the differential inclusion

$$\dot{\theta} \in \Pi_{T_{[0, \pi/2]}(\theta)}(\eta \partial_C(-c)(\theta)),$$

with set-valued projection. Corner points  $\{0, \pi/2\} \cap \Sigma_{\min}$  are handled by intersecting both projections in the exceptional stratum.

**Theorem 3.5.3 ( $\kappa=0$  stratified recovery).** Suppose  $\hat{z} \in C^\omega(\Omega_\theta)$  and  $c_A, c_B \in C^\omega$  near  $\hat{z}(\Omega_\theta)$ . The fixed points of the projected gradient flow (Definition 3.5.2) coincide exactly with the stratified critical points of Definition 3.5.1.

*Proof.* Inside the open interval  $(0, \pi/2) \setminus \Sigma_{\dot{\iota}_{\min}}$ , the tangent cone is  $R$ , so  $\Pi(\eta c') = \eta c'$ , vanishing iff  $c'(\theta^{\dot{\iota}}) = 0$ . At  $\theta^{\dot{\iota}} = 0$ ,  $\Pi_{\dot{\iota}}$ , vanishing iff  $c'_{+\dot{\iota}}(0) \leq 0 \dot{\iota}$ . Symmetrically at  $\pi/2$ . On  $\Sigma_{\min} \cap (0, \pi/2)$ , the differential inclusion vanishes iff  $0 \in \partial_C(-c)$ . Corner points are the exceptional stratum.  $\square$

**Audit result (Paper 12 §3 v2 verification, 2026-05-06).** The 50-seed Paper 9 Monte Carlo (Appendix V.3.i) has been reoptimized at high precision with `scipy.optimize.minimize_scalar` on  $[0, \pi/2]$  at tolerance  $10^{-10}$  rad. All 50/50 seeds satisfy the stratified critical point classification of Definition 3.5.1:

- 3 / 50 smooth interior (Fermat critical points);
- 25 / 50 boundary-left KKT ( $\theta^{\dot{\iota}} = 0$ );
- 3 / 50 boundary-right KKT ( $\theta^{\dot{\iota}} = \pi/2$ );
- 19 / 50  $\Sigma_{\min}$  Clarke generalized-gradient critical.

The seeded symmetric-rate geometry of Paper 9 Proposition 9.5 ( $\theta^{\dot{\iota}} \approx 51^\circ$  on the 20-point grid) reoptimizes to  $\theta^{\dot{\iota}} = 0$ , consistent with the boundary-left-KKT class (Paper 9 v1.3.3 Remark 5.6.1). The non-trivial  $\Sigma_{\min}$  incidence (38% of seeds) motivates the full Filippov treatment with three regimes (3.4.A–C) in Theorem 3.4.1. Verification:  
outbox/paper12/computations/paper12\_q5\_kappa0\_audit.{py,csv} (runtime 0.19 s, deterministic).

**Remark 3.5.4 (The projected flow is forced by the data).** Of the 50 seeded geometries, only 3 admit smooth-interior Fermat critical points; 47 require constraint or stratification handling. Writing the recovery as a classical gradient flow  $\dot{\theta} = \eta c'(\theta)$  without projection or Clarke calculus would miscategorize 94% of the sampled ensemble. The projected-flow framing is required by the empirical landscape, not chosen for mathematical convenience.

### 3.6 The $\kappa > 0$ augmented flow: bounded escape and qualified gain

With  $\kappa > 0$ , the augmented flow inherits the branchwise-analytic regularity of Theorem 3.4.1 and the stratified recovery of Theorem 3.5.3 as the  $\kappa \rightarrow 0$  limit. The new content is the *firing cascade* structure: trajectories in the open interior with NP amplitude exceeding  $N P_{crit}$  undergo discrete jumps along the Schur circle. Boundary-KKT trajectories, where the projection of  $\xi_*$  into the boundary tangent cone vanishes (Remark 3.2.4), do not escape along the linear  $\xi_*$  direction.

**Theorem 3.6.1 (Bounded NP-driven dynamics on  $[0, \pi/2]$ ).** *Let  $\theta^\infty \in [0, \pi/2]$  be a stratified critical point of the unperturbed flow (Theorem 3.5.3). Then for any  $\kappa > 0$ :*

(i) (Interior smooth case.) *If  $\theta^\infty \in (0, \pi/2]$  is a smooth interior critical point with  $N P_\sigma(\theta^\infty) = 0$  for both  $\sigma$ , the augmented flow has  $\theta^\infty$  as a fixed point.*

(ii) (Interior firing case.) *If  $\theta^\infty \in (0, \pi/2]$  is smooth interior with  $N P_\sigma(\theta^\infty) = 1$  for some  $\sigma$  and  $c'_\sigma(\theta^\infty) \neq 0$ , a finite-time firing event transports the trajectory to  $\theta^{+\dot{\iota} = \theta^\infty + \rho_\sigma \cdot \text{sgn}(c'_\sigma(\theta^\infty))} \in [0, \pi/2]_{\dot{\iota}}$ , with the post-jump location interior provided  $\rho_\sigma$  is chosen to respect the Zeno-avoidance condition.*

(iii) (Boundary-KKT case, qualified.) *If  $\theta^\infty \in \{0, \pi/2\}$  is a boundary-KKT critical point, then the projected augmented flow  $\dot{\theta} = \Pi_{T_{[0, \pi/2]}(\theta^\infty)}[-\eta F'_\sigma + \kappa \cdot N P_\sigma \cdot \xi_*]$  vanishes at  $\theta^\infty$  along the linear  $\xi_*$  direction. The boundary state is therefore a fixed point of the augmented flow under the linear NP-pump, and escape requires either (a) higher-order coherence information (Z-axis tie-breaker), or (b) supplementary nonlinear mechanisms beyond the Schur-derived  $\xi_*$ .*

*Proof.* (i) and (ii) are immediate from Definition 3.3.2 and Theorem 3.5.3. (iii) follows from Remark 3.2.4: at  $\theta^\infty = 0$  with  $c'_{+\dot{\iota}(0) \leq 0 \dot{\iota}}$ , the un-projected  $\xi_*^{int} = \text{sgn} \dot{\iota}$ , and  $\Pi_{\dot{\iota} \dot{\iota}}$  projects this to 0. The drift term  $-\eta F'_\sigma = \eta c'_\sigma$  is also projected to 0 under  $\Pi_{\dot{\iota} \dot{\iota}}$  when  $c'_\sigma(0) \leq 0$  (the KKT condition). Mirror argument at  $\pi/2$ .  $\square$

**Audit result (boundary-KKT interior-sup, 2026-05-06).** A direct numerical audit of the 28 boundary-KKT seeds of Theorem 3.5.3

(outbox/paper12/computations/paper12\_q5\_boundary\_kkt\_interior\_sup.{py,csv})

computes  $\dot{\iota}_{\theta \in [10^{-3}, \pi/2 - 10^{-3}]} c(\theta)$  on a 2000-point grid for each seed and compares to  $c(\theta_{boundary}^*)$ .

Findings:

- 24 / 28 boundary-KKT seeds have  $\dot{\iota}_{interior} c < c_{boundary}$ , with a mean shortfall of  $\sim 2.7 \times 10^{-4}$  in  $c$ .
- 4 / 28 have  $\dot{\iota}_{interior} c > c_{boundary}$ , with gains in the range  $[4.2 \times 10^{-4}, 5.1 \times 10^{-2}]$ . These four are concentrated in the boundary-right-KKT class plus a few seeds where the underlying coherence landscape has a non-trivial interior peak that the high-precision optimizer of Theorem 3.5.3 classified as boundary-left only because  $c$  is monotonically decreasing from a local maximum at  $\theta = 0$ .
- The mean gain over boundary-KKT seeds is  $\sim 3.9 \times 10^{-3}$  (skewed positive by the four interior-better cases).

**Implication for Theorem 3.6.1(iii).** For 24 of 28 boundary-KKT seeds, escape from the boundary along *any* mechanism (the un-projected  $\xi_*^{int}$ , or a supplementary nonlinear term as in 3.6.1(iii)(b)) lands the trajectory in coherence-degraded territory. Theorem 3.6.1(iii)’s “escape” is therefore not generically a coherence-improvement mechanism for boundary-KKT geometries; it is at most a *non-trivial trajectory continuation* whose coherence consequences depend on the specific seed. This sharpens the relation to Paper 9 Proposition 9.5: the conditional gain region of Paper 9 is well-defined on a Haar-positive-measure subset of  $(R_A, R_B)$ , but for the 56% of audited seeds at boundary-KKT, the typical escape direction is into lower coherence rather than higher. The four exceptional boundary-KKT seeds with  $\dot{c}_{interior} > c_{boundary}$  are the ones for which the linear  $\xi_*$  mechanism (post-projection into the inward tangent cone after a finite-displacement nonlinear correction) could in principle deliver gain.

**Remark 3.6.2 (Bridge to Paper 11, qualified).** Paper 9 Note 9.6 defers rate improvement to Paper 11 because Schur locks the linear joint rate. Theorem 3.6.1 shows the structural form of any nonlinear escape mechanism (boundary-KKT trajectories require *supplementary* nonlinearity beyond linear  $\xi_*$  to leave the boundary), and the boundary-KKT interior-sup audit shows that even when escape is mechanically possible, *it is generically not a coherence gain*. Paper 11’s analysis must therefore identify the specific non-Schur nonlinear corrections (higher-order multilinear terms, non-equivariant symmetry-breaking) and the specific seed sub-classes for which the post-escape trajectory enters a region of strictly higher  $c$ . **Paper 12 §3 does not establish rate improvement; it provides the dynamical framework, the audited boundary-KKT obstruction, and the residue record within which Paper 11’s rate analysis can be staged.**

### 3.7 The scar record and its role in §4

Each NP firing event indexed by  $i$  on branch  $\sigma_i$  produces a push-forward jump vector

$$v_i := \rho_i \cdot \text{sgn}(c_{\sigma_i}'(\theta_i^-)) \cdot \hat{z}'(\theta_i^-) \in W,$$

recorded in the *physical scar span*

$$S_k := \sum_{i=1}^k R[F_{21}] \cdot v_i \subset W,$$

the *event-indexed scar module*

$$S_k := \bigoplus_{i=1}^k R[F_{21}] \cdot v_i,$$

and the *isotypic multiplicity profile*

$$\mu_k = (\mu_1(S_k), \mu_{L_2}(S_k), \mu_{U_6}(S_k)).$$

Here  $F_{21} \subset G_2$  is the Fano-orientation subgroup inherited from Paper 4; the three real  $F_{21}$ -irreps  $(1, L_2, U_6)$  are characterized in §4.2 (Paper 12 §4 Theorem 4.2.1).

**Remark 3.7.1 ( $F_{21}$ -action on the scar record vs  $F_{21}$ -equivariance of the dynamics).**  $F_{21}$  acts on the ambient  $W$  by the diagonal restriction of the  $G_2$ -action (Paper 9 §3.4 convention), and this action is inherited by every subspace and module of  $W$ . This includes the scar record:  $F_{21}$  acts on  $S_k, S_k, \mu_k$  canonically. The *dynamics* of §3.3, by contrast, is defined for arbitrary  $(R_A, R_B, b_A, b_B) \in O(N) \times O(N) \times R^{2N}$  that need not be  $F_{21}$ -related, so a generic trajectory is *not*  $F_{21}$ -equivariant. The triple  $J_k$  extracts the  $F_{21}$ -content of a generically  $F_{21}$ -asymmetric trajectory; this is the input to §4’s commensurability hierarchy.

### Three structural properties:

- $S_k$  **is saturating**:  $\dim S_k \leq 28 = \dim W$ , so the physical span stabilizes after at most 28 dimension-increasing jumps.
- $S_k$  **is count-faithful as an abstract graded module**: the assignment  $k \mapsto S_k$  is injective by construction. The *natural surjection*  $S_k \twoheadrightarrow S_k$  identifying  $v_i$  with its image in  $W$  has injective kernel iff  $R[F_{21}] \cdot v_i \not\subset S_{i-1}$  for every  $i$  — the generic case under  $c_{\sigma_i}(\theta_i^-) \neq 0$  and  $\rho_i > 0$ .
- $\mu_k$  **is the  $F_{21}$ -type signature** that enables cross- substrate commensurability comparisons (§4).

**Remark 3.7.2 (Representation-theoretic, not topological).** The invariant  $J_k$  is a *representation-theoretic* scar invariant:  $S_k$  is a graded  $F_{21}$ -module,  $\mu_k$  is an isotypic-multiplicity profile, and  $S_k$  is a subspace of an  $F_{21}$ - representation. No topological invariant (winding number, persistent homology) is claimed. Earlier drafts using “topological” language should be read as “representation-theoretic” throughout.

**Remark 3.7.3 (Bridge to §4).** The event-indexed module  $S_k$  is the mathematical realization of the *audit-grounded substrate* principle (Paper 12 thesis Pillar 3): a graded, indexable ledger of coherence adjustments that survives the saturation of the physical span. §4 shows that commensurability of synthetic and biological scar records requires *full  $G_2$ -equivariance* (not merely  $F_{21}$ ) for a canonical cross-substrate isomorphism, with quantitative gap  $\text{End}_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$  vs  $\text{End}_{G_2}(W) \cong M_2(R)$ .

## 3.8 What §3 does and does not establish

### What §3 establishes:

- A Schur-derived canonical tear direction  $\xi$ , inheriting Paper 9’s  $SO(2)$  Schur circle, with line-uniqueness from Schur (Prop 3.2.2) and an explicit coherence-ascent design axiom for the sign (Axiom 3.2.3) plus boundary tangent-cone projection (Remark 3.2.4).
- A branchwise real-analytic augmented flow (Definition 3.3.2) with a five-stratum event set  $\Sigma_{tot}$ , three Filippov regimes (3.4.A–C), and layered Carathéodory/Filippov/hybrid semantics, well- posed on an open dense parameter set  $P^{\dot{c}}$  (Theorem 3.4.1 Conditional Regularity).
- A stratified  $\kappa=0$  recovery theorem (Theorem 3.5.3) that recovers Paper 9’s MC optima as projected-gradient fixed points, with empirical verification via 50-seed audit across all four classes (smooth interior / boundary-KKT /  $\Sigma_{min}$  Clarke).

- A bounded  $\kappa > 0$  augmented-flow theorem (Theorem 3.6.1): interior smooth fixed points fire as expected; boundary-KKT states are *fixed* under linear  $\xi$ , even at  $\kappa > 0$  and require supplementary nonlinearity to escape.
- A boundary-KKT interior-sup audit (24/28 seeds with  $\dot{c}_{\text{interior}} < c_{\text{boundary}}$ ) showing that escape from the boundary is generically *not* a coherence-improvement mechanism. Paper 11's rate analysis must identify specific non-Schur nonlinear corrections.
- The scar invariant triple  $(S_k, S_k, \mu_k)$  as a representation-theoretic ledger of firing history, with explicit distinction between abstract count-faithfulness (always) and the natural-surjection injectivity (generic) (§3.7).

### What §3 does not establish:

- *Rate improvement.* Theorem 3.6.1 shows the structural form of any escape; the boundary-KKT audit shows escape is generically not a gain. Concrete rate analysis is Paper 11's domain (Note 9.6 from Paper 9).
- *A biological mechanism for the  $F_{21}$ -action.* §4's commensurability hierarchy treats biological substrate  $F_{21}$ -content as input, not derived; Paper 13+ would specify the physical mechanism.
- *Cross-substrate commensurability isomorphisms.* §4 territory.
- *A theory of consciousness.* Paper 7's domain.
- *Global well-posedness off  $P^i$ .* Theorem 3.4.1 is conditional on the parameter septuple lying in an open dense set; behavior on the codimension- $\geq 1$  exceptional strata is identified but not analyzed.
- *Higher-multiplicity scar geometries.* §4.4 is restricted to multiplicity-one  $G_2$ -modules; the  $m > 1$  case is deferred.

§4 proceeds to the commensurability hierarchy. The scar invariant  $(S_k, S_k, \mu_k)$  constructed in §3.7 is the object on which §4's equivariant isomorphism theorem acts.

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### References (cited in §3)

[B-2008] M. di Bernardo, C. Budd, A. Champneys, P. Kowalczyk, *Piecewise-smooth Dynamical Systems: Theory and Applications*, Applied Mathematical Sciences 163, Springer (2008). Definitions 3.3.3, 3.3.5 inherit terminology from this reference.

[F-1988] A. Filippov, *Differential Equations with Discontinuous Righthand Sides*, Mathematics and Its Applications 18, Kluwer (1988). Sliding-mode equivalent-control formula (3.4.A) follows §2.7.

[GST-2012] R. Goebel, R. Sanfelice, A. Teel, *Hybrid Dynamical Systems: Modeling, Stability, and Robustness*, Princeton (2012). Hybrid-jump semantics in Definition 3.3.4.

[C-1990] F. Clarke, *Optimization and Nonsmooth Analysis*, SIAM Classics in Applied Mathematics (1990). Generalized-gradient  $\partial_C$  on  $\Sigma_{\min}$  in Definition 3.5.2.

*Verification files referenced:* - outbox/paper12/computations/paper12\_q5\_kappa0\_audit.py (Theorem 3.5.3 audit) - outbox/paper12/computations/paper12\_q5\_kappa0\_audit.csv (per-seed data) - outbox/paper12/computations/paper12\_q5\_boundary\_kkt\_interior\_sup.py (Theorem 3.6.1(iii) boundary audit) - outbox/paper12/computations/paper12\_q5\_boundary\_interior\_sup.csv - outbox/paper12/computations/paper12\_q4\_character\_verify.py (used in §4) - outbox/paper12/chatgpt\_pro\_session\_transcript\_2026-05-05.pdf

*Cross-references:* - Paper 9 v1.3.3 Remark 5.6.1 (grid-snap correction for seed- $k=0$ ) - Paper 9 v1.3.2 Lemma 3.6.1 (Schur uniqueness of  $\Psi_\theta$ ) - Paper 9 v1.3.2 Theorem 9.4 (joint fixed-point existence/analyticity) - Paper 9 v1.3.2 Note 9.6 (rate improvement deferred to Paper 11) - Paper 4 ( $\mathrm{PSL}(2,7) \supset F_{21}$  QBism bridge) - Paper 12 §4 (commensurability hierarchy, in preparation)

*Revision log:* - v2 (2026-05-06 ~01:45 PDT): Council rigor pass applied. R1 load-bearing fix on Theorem 3.6.1 boundary-KKT case, with new audit showing 24/28 boundary seeds have interior coherence below boundary. R2 restated Theorem 3.4.1 as Conditional Regularity with Sard-style open-density claim on parameter set  $P^6$ . Filippov regimes 3.4.A–C made explicit. Branch-wise notation  $Z_\sigma$  + branch-wise derivative usage. Generator  $J$  rendered correctly.  $S_k$  count-faithfulness mis-statement corrected.  $F_{21}$ -equivariance bridge to §4 added (Remark 3.7.1). Pillar-delivery signpost added at top. - v1 (2026-05-06 ~00:30 PDT): First complete draft.